

HydroPulse: Smart Drought Water Supply Chain

MASTER BLUEPRINT & PROJECT ARCHITECTURE | DOMAIN: SMART OPERATIONS & SUSTAINABLE SUPPLY CHAIN

1. Executive Summary & Problem Statement

Recurring summer droughts in semi-arid regions like Marathwada and Ahmednagar create critical drinking water shortages. Traditional water tanker distribution suffers from severe operational bottlenecks: unauthorized diversions (**Tanker Mafia**), ghost billing, and complete neglect of community livestock and stray wildlife.

The Solution: HydroPulse is an end-to-end intelligent platform that predicts regional water scarcity 14 days in advance using satellite telemetry, stops tanker diversion via cryptographic QR validation, and optimizes delivery routes using automated vehicle routing solvers.

2. System Architecture & Tech Stack

Layer 1: Geospatial Engine Function: Tracks soil moisture & surface water levels. Tech: Python, GeoPandas, Rasterio, NASA SMAP, Copernicus API.	Layer 2: Anti-Diversion Module Function: Geofenced QR verification for discharge checkpoints. Tech: Flutter PWA (Offline-First), PostgreSQL.
Layer 3: Logistics Solver Function: Equitable distribution routing ensuring human & cattle quotas. Tech: Google OR-Tools (VRP Solver).	Layer 4: Command Dashboard Function: Real-time distress map for district collectors. Tech: FastAPI, Leaflet.js, Tailwind CSS.

3. Free & Open-Source Data Sources

- **Soil Moisture & Groundwater:** NASA SMAP API & GRACE-FO telemetry.
- **Surface Water & Lakes:** ESA Copernicus Sentinel-2 Open Access API (10m resolution).
- **Rainfall Deficits:** India Meteorological Department (IMD) Gridded Rainfall Data.

Operational Guidelines & Risk Management

MATHEMATICAL MODELING & PRACTICAL MITIGATIONS

4. Mathematical Allocation Framework

To guarantee fairness, water distribution is automated based on statutory distress quotas:

Human Quota: $Q_{\text{human}} = \text{Population} \times 40 \text{ liters/day}$

Cattle Quota: $Q_{\text{cattle}} = \text{Cattle Count} \times 70 \text{ liters/day}$

Total Demand: $D_{\text{total}} = Q_{\text{human}} + Q_{\text{cattle}}$ (Optimized via OR-Tools matrix)

5. Anticipated Operational Challenges & Mitigations

Challenge	Root Cause	Technical & Practical Mitigation
Zero Connectivity	Remote rural hamlets lack stable 2G/4G cellular coverage.	Implement Offline-First PWA architecture . QR handshakes sync locally and upload when network restores.
Tanker Mafia Resistance	Contractors attempting to bypass checkpoints or divert water.	Strict enforcement of " No Cryptographic Proof, No Bill Payment " linked directly to state treasury.
Coarse Satellite Data	Satellite grids cover broad zones rather than single wells.	Use a Hybrid Approach combining satellite telemetry with Central Ground Water Board (CGWB) static stations.
Bureaucratic Inertia	Field staff accustomed to manual paper registers.	Provide a zero-friction, multilingual (Marathi/English) single-tap UI dashboard.

6. Conclusion & Impact

HydroPulse transforms reactive drought relief into an automated, transparent, and predictive water supply chain. By eliminating leakages and prioritizing animal welfare alongside human needs, it ensures efficient resource allocation across rural districts.